SALES PERFORMANCE ANALYSIS REPORT

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Tools Used: Python, Pandas, SQLite, SQL, Power BI, VS Code, Excel

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1. Executive Summary

This project presents an end-to-end Sales Performance Analysis solution developed using Python, SQL, and Power BI. The objective is to transform raw sales transaction data into meaningful business insights that can support strategic decision-making.

The workflow includes data cleaning and preprocessing using Python, business analysis using SQL queries executed on SQLite, and dashboard development using Power BI. The project focuses on analyzing sales trends, regional profitability, customer purchasing patterns, and product performance.

The final dashboard enables stakeholders to monitor key business metrics and identify opportunities for growth and profitability improvement.

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2. Business Problem Statement

Organizations generate large volumes of transactional sales data every day. However, without proper analysis, this data provides limited business value.

The objective of this project is to answer the following business questions:

- Which region generates the highest profit?
- Which products contribute most to revenue?
- Who are the top customers?
- How do sales vary over time?
- What is the overall profit margin of the business?
- Which business areas require improvement?

By answering these questions, management can make informed decisions regarding sales strategy, inventory management, customer retention, and resource allocation.

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3. Project Workflow

The project follows a complete Data Analytics lifecycle:

Raw Excel Dataset
↓
Data Cleaning using Python
↓
Feature Engineering
↓
Store Clean Data in SQLite
↓
SQL Business Analysis
↓
Power BI Dashboard Development
↓
Business Insights & Recommendations

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4. Dataset Summary

Dataset Information

Metric	Value
Original Records	500
Cleaned Records	482
Total Columns	10

Dataset Features

Column Name	Description
Order_ID	Unique transaction ID
Order_Date	Date of order
Customer	Customer name
Region	Sales region
Product	Product purchased
Sales	Revenue generated

Column Name	Description
Cost	Product cost
Profit	Profit earned
Year	Extracted year
Month	Extracted month

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5. Data Cleaning and Transformation Using Python

Data preprocessing was performed using the Pandas library to improve data quality and prepare the dataset for analysis.

Step 1: Import Required Libraries

```
import pandas as pd
```

Step 2: Load Dataset

```
df = pd.read_excel("sales_raw_500.xlsx")
```

Step 3: Remove Duplicate Records

Duplicate transactions can distort business metrics.

```
df = df.drop_duplicates()
```

Step 4: Handle Missing Values

Rows with missing order dates were removed.

```
df = df.dropna(subset=['Order_Date'])
```

Step 5: Feature Engineering

Profit Calculation

```
df['Profit'] = df['Sales'] - df['Cost']
```

Year Extraction

```
df['Year'] = df['Order_Date'].dt.year
```

Month Extraction

```
df['Month'] = df['Order_Date'].dt.month
```

Step 6: Save Clean Dataset

```
df.to_csv("Sales_Cleaned2.csv", index=False)
```

Outcome

- Removed duplicate records.
- Removed invalid date records.
- Created analytical columns.
- Generated a clean dataset for SQL and Power BI.

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6. SQL Database Integration

The cleaned dataset was loaded into a SQLite database.

```
conn = sqlite3.connect("sales.db")

df.to_sql("sales", conn, index=False, if_exists="replace")
```

Benefits:

- Enables SQL analysis.
- Supports structured business reporting.
- Simulates real-world database workflows.

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7. SQL Business Analysis

Query 1: Total Sales, Cost, and Profit

Purpose:
Evaluate overall business performance.

```
SELECT

SUM(Sales) AS Total_Sales,

SUM(Cost) AS Total_Cost,

SUM(Profit) AS Total_Profit

FROM sales;
```

Query 2: Total Sales by Region

Purpose:
Identify revenue-generating regions.

SELECT Region,

SUM(Sales) AS Total_Sales

FROM sales

GROUP BY Region

ORDER BY Total_Sales DESC;

Query 3: Total Profit by Region

Purpose:
Measure regional profitability.

SELECT Region,

SUM(Profit) AS Total_Profit

FROM sales

GROUP BY Region

ORDER BY Total_Profit DESC;

Query 4: Top 10 Customers by Sales

Purpose:
Identify high-value customers.

SELECT Customer,

SUM(Sales) AS Total_Sales

FROM sales

GROUP BY Customer

ORDER BY Total_Sales DESC

LIMIT 10;

Query 5: Top 5 Products by Profit

Purpose:
Identify high-performing products.

```
SELECT Product,

SUM(Profit) AS Total_Profit

FROM sales

GROUP BY Product

ORDER BY Total_Profit DESC

LIMIT 5;
```

Query 6: Average Profit per Order

Purpose:
Measure average profitability.

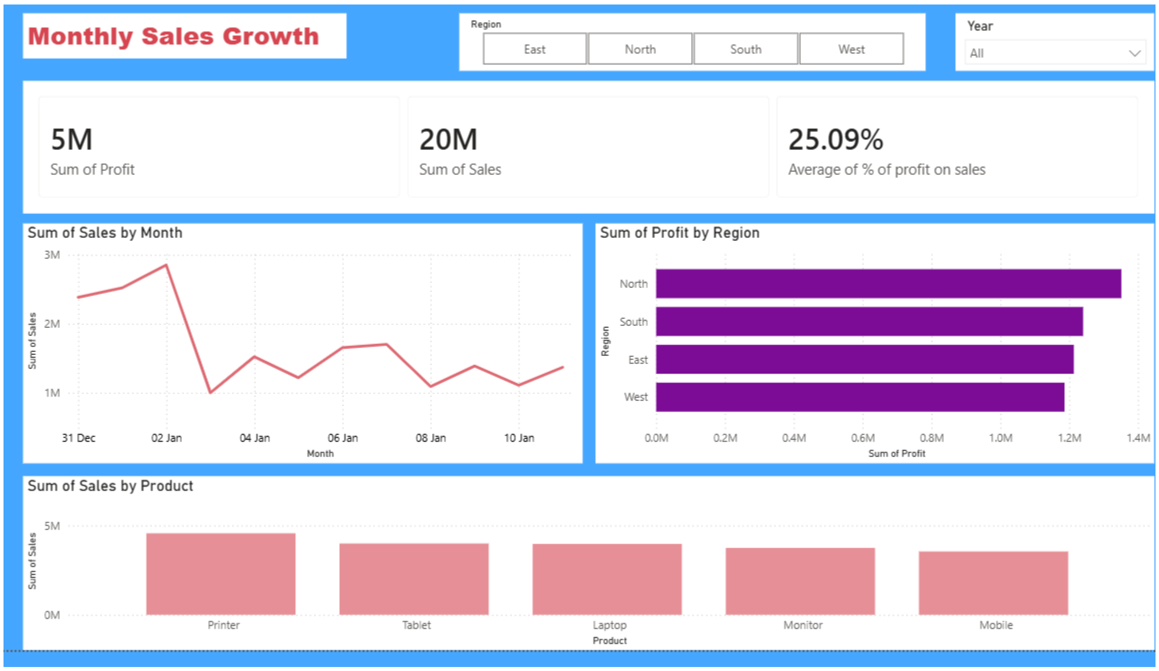
```
SELECT AVG(Profit) AS Avg_Profit

FROM sales;
```

Query 7: Most Profitable Region

Purpose:
Determine highest-profit region.

```
SELECT Region,

SUM(Profit) AS Total_Profit

FROM sales

GROUP BY Region

ORDER BY Total_Profit DESC

LIMIT 1;
```

8. Power BI Dashboard



Dashboard Overview

The Power BI dashboard was created to visualize business performance using interactive charts and KPI cards.

Key Performance Indicators

KPI	Value
Total Sales	20M
Total Profit	5M
Profit Margin	25.09%

Dashboard Visualizations

Monthly Sales Growth

Tracks sales trends and identifies peak sales periods.

Profit by Region

Compares profitability among:

- North
- South
- East
- West

Sales by Product

Compares product performance:

- Printer
- Tablet
- Laptop
- Monitor
- Mobile

Interactive Filters

- Region Filter
- Year Filter

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9. Key Insights

Overall Business Performance

- Total revenue exceeded 20 million.
- Total profit exceeded 5 million.
- Overall profit margin is 25.09%.

Regional Performance

- North region generated the highest profit.
- West region generated the lowest profit.
- Regional performance remains relatively balanced.

Product Performance

- Printer is the highest-selling product.
- Mobile generated the lowest sales.
- Product sales are diversified.

Sales Trends

- Significant sales growth occurred early in the observed period.
- Sales later stabilized.
- Seasonal patterns can be further investigated.

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10. Business Recommendations

Increase Investment in High-Profit Regions

Allocate additional marketing resources to North region.

Improve Underperforming Regions

Conduct market analysis for West region.

Promote Best-Selling Products

Increase inventory and marketing support for top-performing products.

Customer Retention Programs

Implement loyalty and reward initiatives.

Inventory Planning

Use sales trends for forecasting and stock optimization.

Continuous Monitoring

Track KPI performance using the Power BI dashboard.

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11. Project Limitations

- Dataset size is relatively small.
- Analysis focuses on historical data only.

- Customer demographics are unavailable.

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12. Future Improvements

- Build predictive sales forecasting models.
- Develop customer segmentation using Machine Learning.
- Integrate live database connections.
- Automate reporting pipelines using Python.

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13. Conclusion

This project successfully demonstrates an end-to-end Data Analytics workflow involving data cleaning, SQL analysis, and Power BI dashboard development.

The project transformed raw transactional data into actionable business insights and provides a strong foundation for data-driven decision-making.

Technologies Used

- Python
- Pandas
- SQLite
- SQL
- Power BI
- Microsoft Excel
- VS Code
- GitHub